The Integration of the Submissions and the Final Exam

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The Integration of the Submissions

The main challenges remaining:

- Decide what you will use as your navigation control variable (or variables)
- Be sure you can correctly calculate the value of the control variable(s), both without and with intersections
- Adjust your navigation system to the robot's (simulated and/or real) motors and your pyrobot loop speed after incorporating vision to the loop
- Manage failure (lost line, unable to recognize arrow direction, . . .)

Using a Real or a Simulated Robot

Using a real robot for the demo of the final submission is **optional**.

However, the maximum grade of final submissions using a simulated robot will be 7/10. (In both cases the final grade of the class will be calculated as previously mentioned.)

You will need help the first time you use the real robots. We will be available:

- at the end of today's and next week's class
- at the times mentioned in the Wiki in Moodle
  You need to sign up for those sessions. They will be **cancelled** if no one is listed in the Wiki 24 hours before the session.

The *conserjes* have a list of your names and will open the Sala Alpera for you.

You can also use the lab to test homemade robots.

Your Pyrobot Brain with a Real Camera

```
class BrainFinalExam(Brain):
  # ... class variables, etc.

  def setup(self):
    # Put code here which should only be executed ONCE at
    # the start.  For example, connecting to the camera,
    # setting up segmentation etc.  Keep references in
    # instance variables.
    self.capture = cv2.VideoCapture(0)
    # set capture resolution ...

  def destroy(self):
    # Put code here to clean up everything when shutting down.
    cv2.destroyAllWindows()

  def step(self):
    # Put code here which should be executed during every step.
```

The final weeks...

- May 19, 26 - Group help sessions (in the classroom)
- Help sessions in the lab - see Wiki in Moodle
- Individual tutorial sessions with Roberto, Luis or Nik by appointment
- Final Exam: June 4th, 15:00
- Final Exam: July 6th, 10:00

The Final Exam

- Each group will have 15 minutes to demonstrate their work.
- All group members need to be present.
- The robot must be able to:
  - follow the line
  - process Y, T and X intersections and follow the arrows to select the correct branch.
  - recognize and report the location of at least 2 symbols
- The robot should successfully complete the first (known) route. The second ‘surprise’ route will be the same for everyone but will not be announced until the day of the exam.

The Final Exam (cont.)

- Each route can be attempted as many times as you like (during the 15 min.) and the code can also be modified during that time.
- If you have implemented any additional behaviors
  - identify more symbols,
  - search for the line,
  - avoid obstacles,
  - avoid more complex obstacles in more complex locations,
  - ...

  be sure to show them during your time!
- Submit the code that you used for your demo using Moodle.

The Final Exam (cont.)

Real robot - grading of the final project:

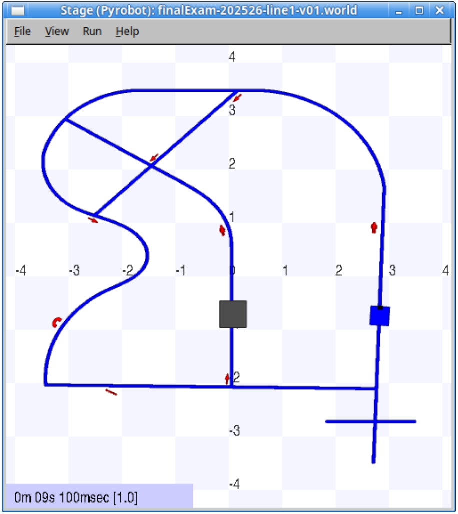
- 5 - Is able to complete the first route once
- up to +1 - Can recognize all 4 markers (most of the time)
- up to +1 - Completes the second route
- up to +1 - Obstacle avoidance
- up to +2 - Moves with confidence and consistency, behaves intelligently, additional behaviors, can follow a ball, etc.

The Final Exam (cont.)

Simulated robot - grading of the final project:

- 4 - Is able to complete the first route once
- up to $+.5$ - Can recognize all 4 markers (most of the time)
- up to $+1$ - Completes the second route
- up to $+.5$ - Obstacle avoidance
- up to $+1$ - Moves with confidence and consistency, behaves intelligently, additional behaviors, etc.

This Year's Line With the First (Known) Route



This Year's Line With the First (Known) Route